HENSELIAN LOCAL RINGS OF A SPACE AND OF ITS BASE CHANGE TO THE COMPLETION HAVE ISOMORPHIC COMPLETIONS

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Editorial note. The point q of |Y| lies over the closed point of Spec(A) by hypothesis; the reduction preserves that correspondence.

Allowed standard prerequisites (editorial).

- Elementary etale neighbourhoods of decent spaces. Decent Spaces, lemma decent-space-elementary-etale-neighbourhood (cross-chapter, named). Provides the etale neighbourhood (U, u) -> (X, p) reducing to the scheme case.
- Completion isomorphism for schemes. Resolution of Surfaces, lemma iso-completions (cross-chapter, named). The scheme case: henselian local rings of U and V = U x_{Spec(A)} Spec(A^) at v have isomorphic completions.

Source reference. Original reference 0BBP: focused-seeds/geometry/S9-spres-iso-completions/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BHP: focused-seeds/geometry/S9-spres-iso-completions/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a local ring with finitely generated maximal ideal $\mathfrak{m}$. Let X be a decent algebraic space over A. Let $Y = X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A^\wedge)$ where $A^\wedge$ is the $\mathfrak{m}$-adic completion of A. For a point $q \in |Y|$ with image $p \in |X|$ lying over the closed point of $\mathrm{Spec}(A)$ the map $\mathcal{O}^h_{X,p} \to \mathcal{O}^h_{Y,q}$ of henselian local rings induces an isomorphism on completions.*

Proof. This follows immediately from the case of schemes by choosing an elementary étale neighbourhood $(U, u) \to (X, p)$ as in Decent Spaces, Lemma 0BBP, setting $V = U \times_X Y = U \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A^\wedge)$ and $v = (u, q)$. The case of schemes is Resolution of Surfaces, Lemma 0BHP. □

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